

CM0845 Logic

First-Order Logic: Natural Deduction

Andrés Sicard-Ramírez

Universidad EAFIT

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Remark

The references for this section are van Dalen (2013, § 3.8 and § 3.9).

Derivation Rules for the Quantifiers

Universal quantifier

$$\frac{\varphi(x)}{\forall x \varphi(x)} \forall I$$

$$\frac{\forall x \varphi(x)}{\varphi[t/x]} \forall E$$

Side condition: In $\forall I$, the variable x may not occur free in any hypothesis on which $\varphi(x)$ depends.

Derivation Rules for the Quantifiers

Existential quantifier

$$\frac{\varphi[t/x]}{\exists x\varphi(x)} \exists\text{I}$$
$$\frac{\begin{array}{c} [\varphi(x)]^x \\ \vdots \\ \exists x\varphi(x) \end{array} \quad \psi}{\psi} \exists\text{E}^x$$

Side condition: In $\exists\text{E}$, the variable x is not free in ψ , or in a hypothesis of the sub-derivation of ψ , other than $\varphi(x)$.

Derivation Rules for the Quantifiers

Example

The derivation rules for the quantifiers are consistent with the convention that the universe of discourse is not empty, so we can prove that $\vdash \forall x\varphi(x) \rightarrow \exists x\varphi(x)$.

Proof

$$\frac{\frac{\frac{[\forall x\varphi(x)]^x}{\varphi(x)} \forall E}{\exists x\varphi(x)} \exists I}{\forall x\varphi(x) \rightarrow \exists x\varphi(x)} \rightarrow I^x$$

Derivation Rules for the Quantifiers

Example

Prove that $\vdash \exists x(\varphi(x) \vee \psi(x)) \rightarrow \exists x\varphi(x) \vee \exists x\psi(x)$ (van Dalen 2013, p. 92).

Proof

$$\frac{\frac{\frac{[\varphi(x)]^x}{\exists x\varphi(x)}\exists\text{I} \quad \frac{[\psi(x)]^y}{\exists x\psi(x)}\exists\text{I}}{\frac{[\varphi(x) \vee \psi(x)]^z}{\exists x\varphi(x) \vee \exists x\psi(x)}\vee\text{I}}\vee\text{I} \quad \frac{[\exists x(\varphi(x) \vee \psi(x))]^w}{\exists x\varphi(x) \vee \exists x\psi(x)}\vee\text{E}^{x,y}}{\frac{\exists x\varphi(x) \vee \exists x\psi(x)}{\exists x(\varphi(x) \vee \psi(x)) \rightarrow \exists x\varphi(x) \vee \exists x\psi(x)}\rightarrow\text{I}^w}\exists\text{E}^z$$

Derivation Rules for the Identity

Identity

$$\frac{}{x = x} \text{RI}_1$$

$$\frac{x = y}{y = x} \text{RI}_2$$

$$\frac{x = y \quad y = z}{x = z} \text{RI}_3$$

$$\frac{x_1 = y_1, \dots, x_n = y_n}{t[x_1, \dots, x_n / z_1, \dots, z_n] = t[y_1, \dots, y_n / z_1, \dots, z_n]} \text{RI}_4$$

$$\frac{x_1 = y_1, \dots, x_n = y_n \quad \varphi[x_1, \dots, x_n / z_1, \dots, z_n]}{\varphi[y_1, \dots, y_n / z_1, \dots, z_n]} \text{RI}_4$$

Derivation Rules for the Identity

Example

The derivation rules for the identity are consistent with the convention that the universe of discourse is not empty, so we can prove that $\vdash \exists x(x = x)$.

Proof

$$\frac{\frac{}{x = x} \text{RI}_1}{\exists x(x = x)} \exists\text{I}$$

References

-  Dirk van Dalen [1980] (2013). Logic and Structure. 5th ed. Springer (cit. on pp. 2, 6).